

SideKick

Design Specification — turning brief evidence into testable success criteria

NAME

CLASS

DATE

THE BRIEF

GLOBAL CONTEXT — SCIENTIFIC & TECHNICAL INNOVATION

SideKick — A Reusable Water Bottle for the School Day

DESIGN PROBLEM

100 words

Many students at the school carry a water bottle to school, but the canteen still sells around 150 single-use plastic bottles every week to students who forgot theirs, left it leaking in their bag, or found their reusable bottle too heavy or awkward to carry. A survey of 40 students found that leaking, bulkiness, and drinks going warm by lunchtime were the three most common complaints about their current bottles. The school now wants to reduce single-use plastic sold at the canteen by giving students a genuine reason to bring, and actually use, a reusable bottle every single school day.

RESEARCH INSIGHTS

A survey of 40 students backed up what canteen staff already suspected: 68% own a reusable bottle, but don't bring it to school every day. Asked why, leaking in the bag and the bottle being "too much effort to carry" came up more than any other reason. Short interviews with three students who do bring a bottle daily found they'd all specifically chosen slim, lightweight bottles that fit an outside pocket rather than the bag's main compartment — bulkier bottles, even good ones, tended to get left at home once term settled in.

Comparing three bottles already in use around school showed a clear trade-off. A flip-top sports bottle was the lightest and easiest to drink from one-handed between classes, but its seal wore out fast, and most owners reported it leaking within a term. A vacuum-insulated steel bottle kept water noticeably cooler through a full school day, which students said mattered most in hot weather, but at nearly double the weight of the others, it was the bottle most likely to get left behind rather than carried.

THE OPPORTUNITY

There's room for a bottle that fits how students already behave, rather than asking them to change: something slim enough for a side pocket, that doesn't leak even when it's tipped upside down inside a moving bag, and that keeps water noticeably cooler than room temperature through a full school day — without the bulk or cost of a vacuum flask.

KEY FEATURES

- ▶ Slim enough to fit a backpack's side pocket, not just the main compartment
- ▶ A seal that stays leak-proof even when the bottle is carried upside down
- ▶ Keeps water noticeably cooler than room temperature by the end of the school day
- ▶ Opens and pours one-handed, without stopping to unscrew a cap
- ▶ Priced low enough for the school to realistically stock and sell at the canteen

THE FORMULA

The solution must / should [testable requirement], verified by [method], because [research finding].

MUST — non-negotiable. If the solution fails this, it hasn't succeeded, no matter what else it gets right.

SHOULD — desirable. It makes the solution better, but missing it isn't fatal.

If you can't imagine testing it, it's not a specification yet.

NEED A NUDGE? FORMATS TO BORROW

Testable requirements look like...

- weighs no more than 250g
- fits inside a 6cm-wide side pocket
- keeps liquid below 15°C for at least 6 hours
- survives a 1.5m drop onto a hard floor
- opens and pours in under 2 seconds, one-handed
- costs no more than \$8 per unit to produce

Verification methods look like...

- a timed trial using a stopwatch
- a measurement with a scale, ruler, or thermometer
- a survey or rating scale completed by a sample of users
- a drop or stress test from a set height/force
- a side-by-side comparison against an existing product
- an independent reviewer checking against a checklist

These are formats to borrow, not answers to copy — none are written for SideKick specifically. Pair a requirement with a method that could genuinely check it.

IF YOU'RE STUCK

function

safety

size & fit

materials

cost

durability

ease of use

user interaction

environment

aesthetics

Not a checklist — a prompt. Only write a criterion if the brief actually gives you evidence to justify it.

YOUR SPECIFICATION – WRITE 6 TO 10 STATEMENTS

1

☐ MUST

☐ SHOULD

The solution must/should ..., verified by ..., because

2

☐ MUST

☐ SHOULD

The solution must/should ..., verified by ..., because

3

☐ MUST

☐ SHOULD

The solution must/should ..., verified by ..., because

4

☐ MUST

☐ SHOULD

The solution must/should ..., verified by ..., because

5

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

6

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

7

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

8

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

9

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

10

☐ MUST ☐ SHOULD

The solution must/should ..., verified by ..., because

SELF - CHECK

- ☐ Each statement is labelled MUST or SHOULD.
- ☐ Every statement is something I could actually imagine testing — a number, a yes/no check, an observable behaviour.
- ☐ Every statement names a realistic verification method.
- ☐ Every statement is traceable to a specific piece of evidence in the brief — not something I've assumed.
- ☐ I haven't just restated the brief's paragraphs — each one is a checkable requirement, not a summary.